

Faculty of Mathematics
and Information Science
Warsaw University of Technology

Group Project

Lecture 1:
Course Setup, Team Dynamics & Generative AI

dr inż. Krzysztof Kaczmarski

2026

Materiały sponsorowane przez:



Politechnika Warszawska



Zadanie 10 pn. „Przygotowanie i uruchomienie nowego kierunku studiów na studiach II stopnia – Inżynieria i Analiza Danych (IAD)”
realizowane jest w ramach projektu
„NERW PW. Nauka — Edukacja — Rozwój — Współpraca”
współfinansowanego ze środków Unii Europejskiej w ramach
Europejskiego Funduszu Społecznego

Goals for today:

- Understanding course objectives, schedule, and defense regulations
- Shifting mindset: from course exercises to an engineering diploma product
- Structuring team dynamics, responsibilities, and avoiding the free-rider trap
- Navigating Generative AI: regulations, ethical use, and mandatory reporting

Lecture 1:
Course Setup, Team Dynamics & Generative AI

- Course Overview & Defense Framework
  - Schedule & Academic Context
  - Lectures
  - Laboratories
  - Application Development
  - Thesis writing
  - Summary
  - Stakeholder Roles
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  - Structuring Team Collaboration
  - The Free-Rider Problem
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  - Regulations & Principles
  - Ethical Engineering with AI
- Work Assignment

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Schedule & Academic Context

Course Overview & Defense Framework

weeks	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
lecture															
laboratories				1	1’	2	2’	3	3’	4	4’	5	5’		
application															
thesis writing															

week 12th	Feature freeze: fully operating, verifiable application
weeks 13–15th	Thesis finalization, supervisor review, polishing
February, 1nd	Strict deadline for uploading thesis to APD system
Winter session	Thesis defense & diploma examination

Remember

The group project is not just another course with a grade. It culminates in your official university diploma thesis defense.

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Lectures

Course Overview & Defense Framework

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- 7-8 weeks $\times$ 2h:
  - Organization
  - Requirements and risk analysis
  - Team work and cooperation
  - User Interface
  - Testing
  - Thesis writing guide
- Test (on the last lecture)
- Max points: 30p.
- Min passing: 15p.

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Laboratories

Course Overview & Defense Framework

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- 5 meetings/stages (every 2 weeks)
- Starting around 4th semester week – see calendar
- Each meeting has got one week of emergency delay
- Each stage gives 8 points
- One week delay -4p (two and more weeks. -8p)
- One delay in a semester with no consequences
- Altogether: 40 points
- Passing grade: ALL STAGES MUST BE COMPLETED (even for 0p)

Laboratories: Stages

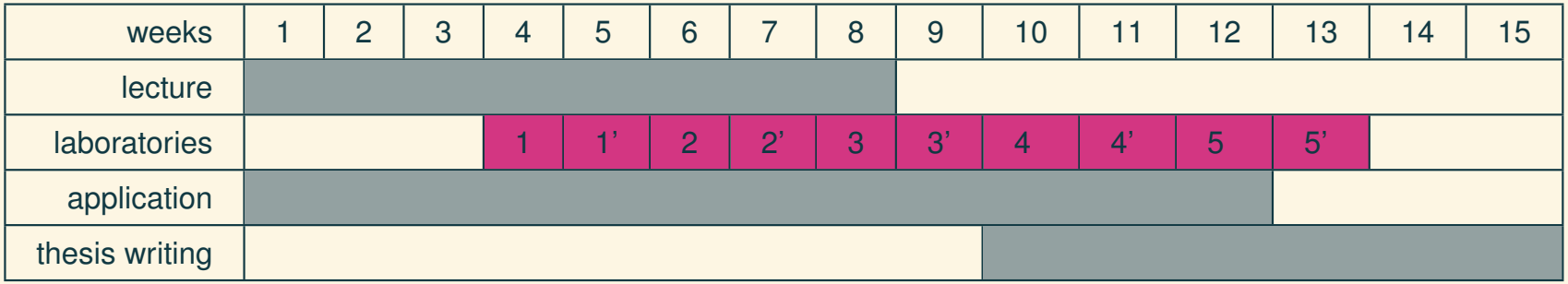
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1. We are ready to begin the project. We know all the problem.
2. We designed the solution and we can start implementation.
3. We are on the way working on the code (correctness).
4. We now start testing
5. We finished testing, the client accepted deployed system.

Laboratories: Artifacts

Course Overview & Defense Framework



There are artifacts required to pass each stage:

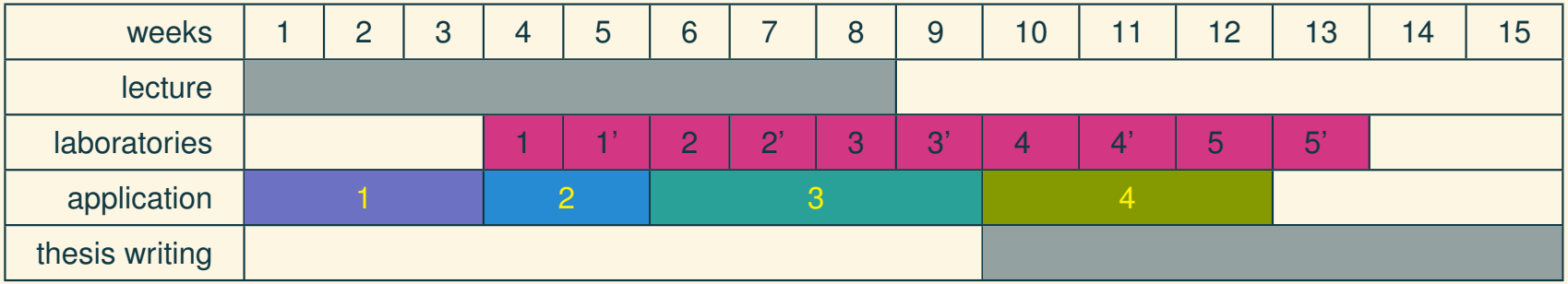
1. Document: business goal, vision of the system, functional and non-functional description, risk analysis, schedule, division of work
2. Document: Problem analysis, design of a solution
3. Code: Main modules implemented (no integration yet)
4. Code: Unit tests
5. Code: All system integrated and tested, Acceptance tests. Document: Deployment documentation (post-completion).

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Application Development

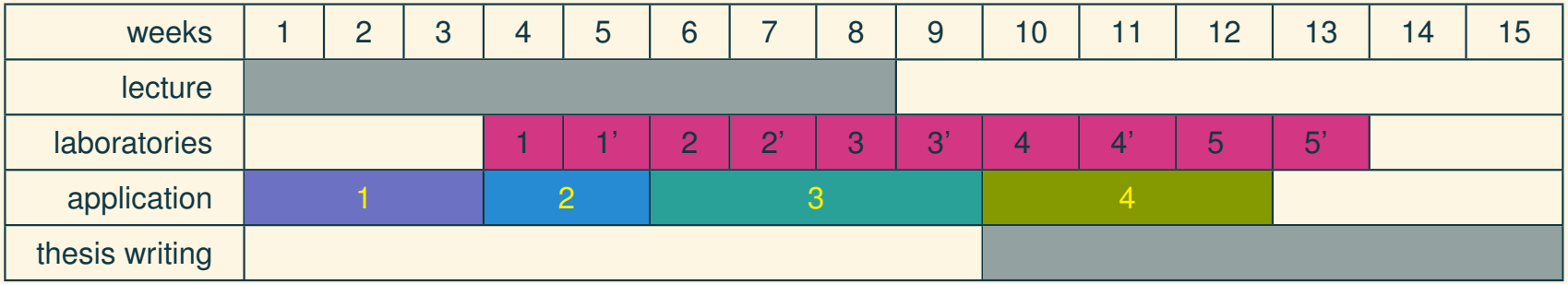
Course Overview & Defense Framework



1. **Inception, Planning** – client requirements, alternative solutions, cost estimation, schedule, risk analysis, feasibility study, use cases
2. **Design** – system architecture, technology selection, modules design, classes design
3. **Implementation** – application, configuration, running reports
4. **Testing, Deployment** – unit tests, acceptance tests, post-deployment documentation, user's guide, installation guide

Application Development

Course Overview & Defense Framework



Requirements

The diploma application must operate in the laboratory of the MiNI Faculty (exceptional situations require the approval of the work supervisor and the consent of the laboratory leaders by the third lab the latest).

- Quality of the final application is graded by the advisor.
- Available points: 0p, 20p, 25p, 30p
- Minimum for passing: 20p

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Work Assignment

Thesis writing

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- The process as well as the detailed content will be explained later during the lecture.

Thesis writing

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NEW REGULATION (from 2024)

- Detailed description of the work done by authors is obligatory.
- Declaration of a scope of generative AI being used is also obligatory.

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Grading Scheme & Passing Criteria

Course Overview & Defense Framework

- Points distribution (max 100 points):
  - **Lecture test / individual contribution:** 30 points
  - **Laboratory stages (5 meetings $\times$ 8 points):** 40 points
  - **Final application quality (graded with supervisor):** 30 points

- Passing thresholds:
  - Lecture: min. 15 points
  - Laboratories: min. 16 points (all 5 stages must be submitted!)
  - Application: min. 20 points
  - Total: min. 51 points for grade 3.0

- **Individualization of grades:** In a group project, team members do not automatically receive the same grade. The supervisor and examiner evaluate individual contributions.

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The Twin-Stakeholder Model

Course Overview & Defense Framework

Laboratory Teacher

Your Project Manager & Quality Inspector

The Twin-Stakeholder Model

Course Overview & Defense Framework

Laboratory Teacher

Your Project Manager & Quality Inspector

Thesis Supervisor (Promotor)

Your Client & Scientific/Domain Advisor

The Twin-Stakeholder Model

Course Overview & Defense Framework

Laboratory Teacher

Your Project Manager & Quality Inspector

Thesis Supervisor (Promotor)

Your Client & Scientific/Domain Advisor

Lab Instructor expects:

- Meeting bi-weekly deadlines
- Formal engineering deliverables
- Code repository & test suite
- Live verification in lab

Supervisor expects:

- Domain problem resolution
- Innovation & scientific rigor
- Acceptance of stage milestones
- High-quality written thesis

Supervisor Acceptance Protocol

Course Overview & Defense Framework

Critical Rule

Before presenting any deliverable at the laboratory meeting, you **must** obtain explicit written or signed acceptance from your supervisor.

- Schedule regular syncs (e.g. bi-weekly 30-min meetings) with your supervisor.

Supervisor Acceptance Protocol

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- Send concise meeting minutes immediately after every consultation.
- Do not surprise your supervisor with a 60-page document two days before the deadline.

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- Schedule regular syncs (e.g. bi-weekly 30-min meetings) with your supervisor.
- Send concise meeting minutes immediately after every consultation.
- Do not surprise your supervisor with a 60-page document two days before the deadline.
- If your supervisor travels or is unavailable, plan ahead and notify the lab teacher in advance.

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Why Group Diploma Projects Fail

Team Dynamics & Governance

Common fatal failure modes:

1. **Big-bang integration**: working in silos until week 11.
2. **Unclear ownership**: "everyone is responsible" means no one is.
3. **Free-rider effect**: one member ghosts the team.
4. **Unchecked scope creep**: endless supervisor ideas with no MVP boundary.

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The Antidote

- Explicit contract and work division
- Weekly sprint syncs (even 15 min)
- Continuous repository commits
- Documented task allocation

Responsibility Matrix (RACI)

Team Dynamics & Governance

Task / Deliverable	Alice	Bob	Carol	David
System Architecture Design	A/R	C	C	I
Core API / Backend Engine	C	A/R	I	R
Frontend / UI Dashboard	I	I	A/R	C
Data Pipeline / Model Training	R	C	I	A/R
Test Automation & CI Setup	R	A/R	I	I
Deployment (Docker / Lab)	A/R	R	C	I
Thesis Chapter 3 (Solution)	A/R	R	R	R

R = Responsible (does the work),
A = Accountable (ensures delivery),
C = Consulted,
I = Informed. Exactly one **A** per task!

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Addressing the Free-Rider Issue Early

Team Dynamics & Governance

The Reality

In almost every semester, at least one group faces a student who stops responding, misses deadlines, or submits unacceptable quality code.

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Never wait until week 12!

If you report a non-contributing member during the final week,

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The 2024+ University Regulations on GenAI

Generative AI in Diploma Projects

Warsaw University of Technology Formal Policy

Using generative AI tools (LLMs, code generation assistants) is permitted **only** when fully disclosed, documented, and ethically aligned with academic integrity.

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  - Prompts or methodology used to verify AI-generated outputs.

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Acceptable vs. Unacceptable Use

Generative AI in Diploma Projects

Acceptable Use

- Brainstorming architecture alternatives
- Explaining obscure library bugs
- Generating repetitive test mocks
- Improving English grammar/style in drafts
- Learning new frameworks faster

Unacceptable Use

- Generating entire thesis sections without comprehension
- Uncritical copy-pasting of hallucinated citations
- Leaking proprietary or sensitive client data into public clouds
- Claiming AI-generated ideas as personal innovation

The "Exam Defense" Test for AI Code

Generative AI in Diploma Projects

Golden Rule of AI-Assisted Development

If you cannot explain every line of code, every architectural choice, and every formula on the blackboard during your diploma exam, **you do not own that code**.

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- Examination committee members will intentionally ask you to explain subtleties in your submitted code.
- Stating *"I don't know, Copilot generated this"* will immediately jeopardize your exam.
- Use AI to amplify your productivity, not to substitute your engineering judgment.

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Focus Group: Team Setup

Work Assignment

Team Exercise

- **Agree on team communication channels (Slack/Discord/Matrix, issue tracker).**
- **Draft your initial RACI matrix for the 5 laboratory stages.**
- **Schedule your recurring bi-weekly consultation with your supervisor.**
- **Choose your version control platform (GitHub / GitLab) and create the project repo.**

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